The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Yocto-Section-Minimizer-Section-Generating Substrate Selector Dynamics
from Selector-Generating Substrate Pre-Selector Dynamics

Abstract

The current primitive-reservoir observer-reduction frontier already sharpened the live burden below the observer-relevant yocto-section minimizer-section core itself: the current yocto-section-minimizer-section-generating substrate selector dynamics package was shown to canonically generate that section core. After that theorem, another same-output deeper-origin relay that merely preserves the same selector dynamics no longer counts as the honest next benchmark-facing move. The next admissible task is therefore to derive or justify that selector-dynamics package itself from more primitive substrate structure, or else prove a still smaller collapse strictly inside it.

The present note takes the direct derivation route. For each sector it introduces a selector-generating substrate pre-selector dynamics package consisting of a compact convex pre-selector body over the already fixed selector body, an affine projection to that fixed body, a nonnegative smooth pre-selector functional with unique fiberwise minimizers, and an exact pre-selector shell projecting diffeomorphically to the recorded exact selector shell. The current selector functional is then no longer an independent primitive stopping point. It is the effective fiberwise minimum value induced by that deeper pre-selector law, and the current exact selector shell is the projected exact shell of the deeper package.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore a genuine benchmark-facing gain below the current selector-dynamics frontier: the remaining honest burden is no longer the current yocto-section-minimizer-section-generating substrate selector dynamics as such, but the deeper selector-generating substrate pre-selector dynamics that canonically generate them, or a still smaller collapse strictly inside that deeper package.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again reached a sharper burden than another same-output relay. The current active-primary theorem already showed that the current observer-relevant yocto-section minimizer-section core is not primitive: it is the canonical output of a deeper yocto-section-minimizer-section-generating substrate selector dynamics package $\mathfrak{C}^{\mathrm{yoc,sel}}$. The routing consequence is then immediate. If the project goal is to keep moving the same benchmark one-metric package toward a disciplined first-principles recovery, the next honest benchmark-facing manuscript must either derive or justify $\mathfrak{C}^{\mathrm{yoc,sel}}$ itself from still more

primitive substrate structure, or prove a genuinely smaller theorem-level collapse strictly inside that package.

The present note takes the direct derivation route. Its claim is not that final substrate microphysics has been identified, and it is not that the benchmark exact-relativistic package has changed. Its claim is narrower and benchmark facing. The current selector package is shown to arise canonically from a more primitive pre-selector layer whose projected exact-shell transport already contains the benchmark-relevant structure of $\mathfrak{C}^{\mathrm{yoc,sel}}$. In that precise sense the current selector dynamics are no longer the primitive place where the active line has to stop.

Two features matter here. First, the theorem targets the current selector package itself rather than re-spending the already settled yocto-section minimizer-section-core handoff, so it does not count progress merely by reproducing the same current section core from yet another relay. Second, the deeper pre-selector package is not merely a renaming of the current selector package. It adds an explicit pre-selector state space, a pre-selector body, a pre-selector-to-selector projection, and a deeper fiberwise minimizing law whose effective energy reproduces the recorded selector functional on the benchmark-relevant body.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not alter the benchmark exact-closure package. It does not introduce a second operative metric or a new low-energy matter law. It only proves that the current yocto-section-minimizer-section-generating substrate selector dynamics can itself be generated from a more primitive pre-selector-level substrate dynamics.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded zepto-section benchmark base)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already fixes the following benchmark-facing zepto-section base data:*

1. *a finite-dimensional Euclidean zepto-section state space*

$$\mathcal{U}^{\mathrm{zeptosec}}_\sigma;$$

2. *a compact convex zepto-section body*

$$\mathcal{B}^{\mathrm{zeptosec}}_\sigma \subset \mathcal{U}^{\mathrm{zeptosec}}_\sigma$$

with nonempty interior;

3. *a nonnegative smooth zepto-section functional*

$$\mathscr{U}^{\mathrm{zeptosec}}_\sigma : \mathcal{U}^{\mathrm{zeptosec}}_\sigma \to [0, \infty);$$

4. *a closed embedded exact zepto-section shell*

$$\mathcal{Z}^{\mathrm{zeptosec}}_\sigma \subset \mathcal{B}^{\mathrm{zeptosec}}_\sigma;$$

5. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and use of this zepto-section base as already fixed upstream data on the active line.*

Definition 3 (Recorded downstream chain below the current selector frontier)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already fixes the following downstream data and consequences:*

1. *the current observer-relevant yocto-section minimizer-section core* $\mathfrak{C}^{\mathrm{yoc,sec}}$*, the current observer-relevant yocto-section minimizer-image core* $\mathfrak{C}^{\mathrm{yoc,img}}$*, the current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, the current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, the current micro-section package* $\mathfrak{W}^{\mathrm{microsec}}$*, the current sub-section package* $\mathfrak{V}^{\mathrm{subsec}}$*, and the previously recorded shell-transport consequences used on the active line;*

2. *the previously recorded fact that, once the current yocto-section-minimizer-section-generating substrate selector dynamics package is available, the current observer-relevant yocto-section minimizer-section core, the current observer-relevant yocto-section minimizer-image core, and all of the downstream data listed in item (1) follow unchanged.*

Definition 4 (Recorded yocto-section-minimizer-section-generating substrate selector dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The recorded current selector dynamics package* $\mathfrak{C}^{\mathrm{yoc,sel}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean selector state space* $\mathcal{D}_\sigma^{\mathrm{yoc,sel}}$*, a compact convex selector body*

$$\mathcal{B}_\sigma^{\mathrm{yoc,sel}} \subset \mathcal{D}_\sigma^{\mathrm{yoc,sel}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{yoc,sel}} : \mathcal{D}_\sigma^{\mathrm{yoc,sel}} \to \mathcal{B}_\sigma^{\mathrm{zeptosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{yoc,sel}}(\mathcal{B}_\sigma^{\mathrm{yoc,sel}}) = \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

For each $q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}$*, write*

$$\mathcal{F}_\sigma^{\mathrm{yoc,sel}}(q) \equiv (\Pi_\sigma^{\mathrm{yoc,sel}})^{-1}(q) \cap \mathcal{B}_\sigma^{\mathrm{yoc,sel}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth selector functional*

$$\mathscr{L}_\sigma^{\mathrm{yoc,sel}} : \mathcal{D}_\sigma^{\mathrm{yoc,sel}} \to [0, \infty)$$

such that, for every $q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}$*, the restriction of* $\mathscr{L}_\sigma^{\mathrm{yoc,sel}}$ *to* $\mathcal{F}_\sigma^{\mathrm{yoc,sel}}(q)$ *is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on* q*.*

3. *A closed embedded exact selector shell*

$$\mathcal{Z}_\sigma^{\mathrm{yoc,sel}} \subset \mathcal{B}_\sigma^{\mathrm{yoc,sel}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{yoc,sel}} = \left\{ y \in \mathcal{B}_\sigma^{\mathrm{yoc,sel}} : \begin{array}{l} \Pi_\sigma^{\mathrm{yoc,sel}}(y) \in \mathcal{Z}_\sigma^{\mathrm{zeptosec}}, \\ \mathscr{L}_\sigma^{\mathrm{yoc,sel}}(y) = \displaystyle\min_{z \in \mathcal{F}_\sigma^{\mathrm{yoc,sel}}(\Pi_\sigma^{\mathrm{yoc,sel}}(y))} \mathscr{L}_\sigma^{\mathrm{yoc,sel}}(z) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{yoc,sel}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoc,sel}}} : \mathcal{Z}_\sigma^{\mathrm{yoc,sel}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as the fixed benchmark-relevant selector package used on the active line.*

Remark 1. *Definitions 1–4 are fixed inputs. The one operative metric, the ordinary matter route, the recorded selector body* $\mathcal{B}_\sigma^{\mathrm{yoc,sel}}$*, the selector functional* $\mathscr{L}_\sigma^{\mathrm{yoc,sel}}$*, the exact selector shell* $\mathcal{Z}_\sigma^{\mathrm{yoc,sel}}$*, and the already recorded downstream consequences are not reopened here. The present note asks only whether the current selector package can itself be generated from still more primitive substrate structure.*

3. Selector-Generating Substrate Pre-Selector Dynamics

Definition 5 (Selector-generating substrate pre-selector dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* selector-generating substrate pre-selector dynamics package $\mathfrak{D}^{\mathrm{presel}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean pre-selector state space* $\mathcal{E}_\sigma^{\mathrm{presel}}$*, a compact convex pre-selector body*
$$\mathcal{B}_\sigma^{\mathrm{presel}} \subset \mathcal{E}_\sigma^{\mathrm{presel}}$$
with nonempty interior, and an affine surjection
$$\Pi_\sigma^{\mathrm{presel}} : \mathcal{E}_\sigma^{\mathrm{presel}} \to \mathcal{D}_\sigma^{\mathrm{yoc,sel}}$$
satisfying
$$\Pi_\sigma^{\mathrm{presel}}(\mathcal{B}_\sigma^{\mathrm{presel}}) = \mathcal{B}_\sigma^{\mathrm{yoc,sel}}.$$
For each $y \in \mathcal{B}_\sigma^{\mathrm{yoc,sel}}$*, write*
$$\mathcal{F}_\sigma^{\mathrm{presel}}(y) \equiv (\Pi_\sigma^{\mathrm{presel}})^{-1}(y) \cap \mathcal{B}_\sigma^{\mathrm{presel}}.$$
Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth pre-selector functional*
$$\mathscr{P}_\sigma^{\mathrm{presel}} : \mathcal{E}_\sigma^{\mathrm{presel}} \to [0, \infty)$$
such that, for every $y \in \mathcal{B}_\sigma^{\mathrm{yoc,sel}}$*, the restriction of* $\mathscr{P}_\sigma^{\mathrm{presel}}$ *to* $\mathcal{F}_\sigma^{\mathrm{presel}}(y)$ *is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on* y*; write it as*
$$\mathfrak{p}_\sigma^{\mathrm{presel,sel}} : \mathcal{B}_\sigma^{\mathrm{yoc,sel}} \to \mathcal{B}_\sigma^{\mathrm{presel}}.$$
These data satisfy
$$\Pi_\sigma^{\mathrm{presel}} \circ \mathfrak{p}_\sigma^{\mathrm{presel,sel}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{yoc,sel}}},$$
and the effective minimum value recovers the recorded selector functional:
$$\mathscr{L}_\sigma^{\mathrm{yoc,sel}}(y) = \mathscr{P}_\sigma^{\mathrm{presel}}(\mathfrak{p}_\sigma^{\mathrm{presel,sel}}(y)) = \min_{u \in \mathcal{F}_\sigma^{\mathrm{presel}}(y)} \mathscr{P}_\sigma^{\mathrm{presel}}(u) \qquad \text{for every } y \in \mathcal{B}_\sigma^{\mathrm{yoc,sel}}.$$

3. *A closed embedded exact pre-selector shell*
$$\mathcal{Z}_\sigma^{\mathrm{presel}} \subset \mathcal{B}_\sigma^{\mathrm{presel}}$$
satisfying
$$\mathcal{Z}_\sigma^{\mathrm{presel}} = \left\{ u \in \mathcal{B}_\sigma^{\mathrm{presel}} : \begin{array}{l} \Pi_\sigma^{\mathrm{presel}}(u) \in \mathcal{Z}_\sigma^{\mathrm{yoc,sel}}, \\ \mathscr{P}_\sigma^{\mathrm{presel}}(u) = \min\limits_{v \in \mathcal{F}_\sigma^{\mathrm{presel}}(\Pi_\sigma^{\mathrm{presel}}(u))} \mathscr{P}_\sigma^{\mathrm{presel}}(v) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{presel}}\big|_{\mathcal{Z}_\sigma^{\mathrm{presel}}} : \mathcal{Z}_\sigma^{\mathrm{presel}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{yoc,sel}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as a deeper substrate-side source for the current selector package rather than as a replacement benchmark theory.*

Remark 2. *The label* pre-selector *is used here only as a local marker for a deeper layer below the current selector-dynamics frontier. It does not by itself assert a completed final selector ontology or a finished primitive law. Its role is only to name the next sharper package below the current selector dynamics.*

4. Canonical Recovery of the Current Selector Package

Proposition 1 (The canonical pre-selector minimizer section is well defined)**.** *For every benchmark-faithful pre-selector dynamics package, the map* $\mathfrak{p}_\sigma^{\mathrm{presel,sel}}$ *of Definition 5 is a smooth section of* $\Pi_\sigma^{\mathrm{presel}}$*, so that*

$$\Pi_\sigma^{\mathrm{presel}} \circ \mathfrak{p}_\sigma^{\mathrm{presel,sel}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{yoc,sel}}}.$$

Proof. Definition 5(1) makes each fiber $\mathcal{F}_\sigma^{\mathrm{presel}}(y)$ nonempty, compact, and convex. Definition 5(2) states that $\mathscr{P}_\sigma^{\mathrm{presel}}$ has a unique minimizer on each such fiber and that the minimizer depends smoothly on y. Definition 5(2) therefore gives a well-defined smooth map

$$\mathfrak{p}_\sigma^{\mathrm{presel,sel}} : \mathcal{B}_\sigma^{\mathrm{yoc,sel}} \to \mathcal{B}_\sigma^{\mathrm{presel}}.$$

Because every minimizer lies in the fiber over the point at which it is selected, we have

$$\Pi_\sigma^{\mathrm{presel}}\big(\mathfrak{p}_\sigma^{\mathrm{presel,sel}}(y)\big) = y$$

for every $y \in \mathcal{B}_\sigma^{\mathrm{yoc,sel}}$, which is the displayed identity. □

Proposition 2 (The exact selector shell is the projected exact pre-selector shell)**.** *For every benchmark-faithful pre-selector dynamics package,*

$$\Pi_\sigma^{\mathrm{presel}}\big(\mathcal{Z}_\sigma^{\mathrm{presel}}\big) = \mathcal{Z}_\sigma^{\mathrm{yoc,sel}},$$

and the shell restriction

$$\Pi_\sigma^{\mathrm{presel}}\big|_{\mathcal{Z}_\sigma^{\mathrm{presel}}} : \mathcal{Z}_\sigma^{\mathrm{presel}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{yoc,sel}}$$

is a diffeomorphism with inverse

$$\mathfrak{p}_\sigma^{\mathrm{presel,sel}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoc,sel}}} : \mathcal{Z}_\sigma^{\mathrm{yoc,sel}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{presel}}.$$

Proof. Definition 5(3) states directly that

$$\Pi_\sigma^{\mathrm{presel}}\big|_{\mathcal{Z}_\sigma^{\mathrm{presel}}} : \mathcal{Z}_\sigma^{\mathrm{presel}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{yoc,sel}}$$

is a diffeomorphism, so the image identity follows immediately. To identify the inverse, let $y \in \mathcal{Z}_\sigma^{\mathrm{yoc,sel}}$. By Proposition 1, $\mathfrak{p}_\sigma^{\mathrm{presel,sel}}(y)$ lies in the fiber over y. Since $y \in \mathcal{Z}_\sigma^{\mathrm{yoc,sel}}$, Definition 5(3) shows that this point belongs to $\mathcal{Z}_\sigma^{\mathrm{presel}}$. Hence the shell restriction of $\mathfrak{p}_\sigma^{\mathrm{presel,sel}}$ is indeed the inverse diffeomorphism. □

Proposition 3 (The induced selector package is canonical)**.** *Fix a benchmark-faithful pre-selector dynamics package. If*

$$s : \mathcal{B}_\sigma^{\mathrm{yoc,sel}} \to \mathcal{B}_\sigma^{\mathrm{presel}}$$

is any smooth section satisfying

$$\Pi_\sigma^{\mathrm{presel}} \circ s = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{yoc,sel}}}$$

and

$$\mathscr{P}_\sigma^{\mathrm{presel}}\big(s(y)\big) = \min_{u \in \mathcal{F}_\sigma^{\mathrm{presel}}(y)} \mathscr{P}_\sigma^{\mathrm{presel}}(u) \qquad \textit{for every } y \in \mathcal{B}_\sigma^{\mathrm{yoc,sel}},$$

then $s = \mathfrak{p}_\sigma^{\mathrm{presel,sel}}$. In particular, the induced current selector package carries no residual pre-selector-fiber minimizing-choice freedom once the deeper pre-selector dynamics are fixed.

Proof. For each $y \in \mathcal{B}_\sigma^{\mathrm{yoc,sel}}$, the point $s(y)$ belongs to $\mathcal{F}_\sigma^{\mathrm{presel}}(y)$ and minimizes $\mathscr{P}_\sigma^{\mathrm{presel}}$ there. By Definition 5(2), that minimizer is unique. Therefore

$$s(y) = \mathfrak{p}_\sigma^{\mathrm{presel,sel}}(y)$$

for every y, hence $s = \mathfrak{p}_\sigma^{\mathrm{presel,sel}}$. □

Proposition 4 (All current downstream uses factor through the induced selector package)**.** *For every benchmark-faithful pre-selector dynamics package, all current benchmark-facing downstream uses of the current selector package $\mathfrak{C}^{\mathrm{yoc,sel}}$ on the active line factor through the induced selector data together with the fixed pre-selector package. In particular, the already recorded recovery of the current observer-relevant yocto-section minimizer-section core $\mathfrak{C}^{\mathrm{yoc,sec}}$, the current observer-relevant yocto-section minimizer-image core $\mathfrak{C}^{\mathrm{yoc,img}}$, the current observer-relevant zepto-section minimizer-section core $\mathfrak{B}^{\mathrm{zep,sec}}$, the current observer-relevant zepto-section minimizer-image core $\mathfrak{B}^{\mathrm{zep,img}}$, the current atto-section package $\mathfrak{A}^{\mathrm{attosec}}$, the current femto-section package $\mathfrak{Z}^{\mathrm{femtosec}}$, the current pico-section package $\mathfrak{Y}^{\mathrm{picosec}}$, the current observer-relevant pico-section minimizer-section core $\mathfrak{Y}^{\mathrm{pic,sec}}$, the current nano-section package $\mathfrak{X}^{\mathrm{nanosec}}$, the current micro-section package $\mathfrak{W}^{\mathrm{microsec}}$, the current sub-section package $\mathfrak{V}^{\mathrm{subsec}}$, and the previously recorded shell-transport consequences do not require any additional primitive selector datum once the induced selector package is fixed.*

Proof. By Definition 5(1)–(3), the deeper pre-selector package already supplies the recorded selector body, selector functional, and exact selector shell. Proposition 1 gives the canonical pre-selector minimizer section over that body, and Proposition 2 gives the exact shell projection and inverse lift. Definition 3(2) then records that, once the current selector package is available, the current observer-relevant yocto-section minimizer-section core, the current observer-relevant yocto-section minimizer-image core, and all of the downstream data and consequences listed there follow unchanged. Therefore all current benchmark-facing downstream uses at selector scope already factor through the induced selector package together with the fixed pre-selector data, and no additional primitive selector datum is required. □

Proposition 5 (The benchmark package remains fixed)**.** *Every benchmark-faithful selector-generating substrate pre-selector dynamics package preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 5(4). The present note only derives the current selector package from deeper substrate pre-selector data. It does not alter the benchmark exact-closure package. □

5. Main Theorem

Theorem 1 (Yocto-section-minimizer-section-generating substrate selector dynamics from selector-generating substrate pre-selector dynamics)**.** *Fix the primitive continuation data of Definition 1, the recorded zepto-section benchmark base of Definition 2, the recorded downstream chain of Definition 3, the recorded current selector dynamics of Definition 4, and a benchmark-faithful selector-generating substrate pre-selector dynamics package in the sense of Definition 5. Then, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the current yocto-section-minimizer-section-generating substrate selector dynamics are no longer an unexplained primitive stopping point on the active line: they are recovered canonically from the deeper pre-selector dynamics. More precisely:*

1. *the deeper pre-selector dynamics generate the current selector body, the canonical pre-selector minimizer lift over that body, the effective selector functional, and the exact selector shell by Definition 5, Proposition 1, and Proposition 2;*
2. *the induced current selector package is canonical and carries no residual pre-selector-fiber minimizing-choice freedom once the deeper package is fixed by Proposition 3;*
3. *all current benchmark-facing downstream uses of the current selector package already factor through the induced datum by Proposition 4;*
4. *the benchmark boundary remains fixed by Proposition 5;*
5. *consequently the remaining honest burden below the current selector-dynamics frontier is no longer the current yocto-section-minimizer-section-generating substrate selector dynamics $\mathfrak{C}^{\mathrm{yoc,sel}}$ as such. What remains is to derive or justify the deeper selector-generating substrate pre-selector dynamics $\mathfrak{D}^{\mathrm{presel}}$ from still more primitive substrate structure, or else prove a still sharper theorem-level collapse strictly inside that deeper package.*

Proof. Item (1) is the content of Definition 5 together with Propositions 1 and 2. Item (2) is Proposition 3. Item (3) is Proposition 4. Item (4) is Proposition 5. Item (5) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the selector-generating substrate pre-selector dynamics $\mathfrak{D}^{\mathrm{presel}}$ from still more primitive substrate structure in a way that changes benchmark-facing status;*

2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{D}^{\mathrm{presel}}$ *and prove a sharper collapse to it;*

3. *do either of the previous two while preserving the same benchmark one-metric package and the same claim boundary.*

Another same-output deeper-origin relay that merely preserves the current pre-selector package $\mathfrak{D}^{\mathrm{presel}}$ *and therefore merely reproduces the same current selector package* $\mathfrak{C}^{\mathrm{yoc,sel}}$*, the same current observer-relevant yocto-section minimizer-section core* $\mathfrak{C}^{\mathrm{yoc,sec}}$*, the same current observer-relevant yocto-section minimizer-image core* $\mathfrak{C}^{\mathrm{yoc,img}}$*, the same current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the same current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, and the same downstream observer-relevant atto-section, femto-section, pico-section, nano-section, micro-section, and sub-section consequences, another re-expansion back to the larger yocto-section minimizer-image-core, yocto-section minimizer-section-core, or full yocto-section presentations as if they were still independently live, another reopening of larger screened layers, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden. If a quantum continuation is pursued, it matters benchmark-facingly only insofar as it derives this same deeper pre-selector dynamics, collapses them further, or closes a benchmark derivation gate in a genuinely new way.*

Proof. Theorem 1 shows that the current selector package is no longer the deepest unexplained datum on the active line. Therefore a subsequent manuscript must improve on the deeper pre-selector dynamics rather than merely restate the already generated selector package. □

6. Conclusion

The positive result of this note is that the current yocto-section-minimizer-section-generating substrate selector dynamics are no longer primitive on the active derivational line. Once one works with a selector-generating substrate pre-selector dynamics package over the already fixed selector body, the current selector functional and exact selector shell are forced as the effective fiberwise output of a deeper substrate selection law. All current downstream benchmark-facing uses already factor through that induced selector package together with the fixed pre-selector package.

This preserves the same benchmark one-metric package and the same claim boundary. The result does not yet provide a completed first-principles derivation of GR from substrate microphysics, and it does not license stronger promotion language. Its narrower benchmark-facing gain is that the remaining honest burden now sits one level lower again: not on the current selector package itself, but on the deeper pre-selector dynamics that canonically generate it, or on a still smaller collapse strictly inside that deeper package.